

**Feature Importance Analysis for Purchase Prediction Using XGBoost and
LightGBM**

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

Name N.Teja sri roll 2520030139

Name B.Hema sri roll 2520030159

Under the Esteemed guidance of

DR.Shaik Asif

<Assistant Professor, CSE>

<COMPUTER SCIENCE AND ENGINEERING>



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



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DEPARTMENT OF

Name of Title: Feature Importance Analysis for Purchase Prediction Using XGBoost and LightGBM

1. Abstract:

The rapid growth of e-commerce platforms has generated massive amounts of customer data, making purchase prediction an important application of Machine Learning. Predicting whether a customer is likely to purchase a product enables businesses to improve marketing strategies, enhance customer engagement, and increase sales. This project focuses on developing an intelligent purchase prediction model using two powerful gradient boosting algorithms, XGBoost (Extreme Gradient Boosting) and LightGBM (Light Gradient Boosting Machine). These algorithms are well-known for their high prediction accuracy, computational efficiency, and ability to handle large-scale datasets.

The project begins with data preprocessing, which includes handling missing values, encoding categorical attributes, removing duplicate records, and normalizing numerical features where necessary. The cleaned dataset is then divided into training and testing sets for model development. Both XGBoost and LightGBM models are trained using the prepared data, and their performances are evaluated using Machine Learning evaluation metrics such as Accuracy, Precision, Recall, F1-Score, ROC-AUC Score, and Confusion Matrix.

A significant aspect of this project is Feature Importance Analysis, which identifies the most influential factors affecting customer purchase decisions. Features such as product category, customer demographics, browsing history, previous purchase behavior, product ratings, discounts, and session duration are analyzed to determine their contribution to prediction accuracy. Comparing feature importance from both XGBoost and LightGBM provides valuable insights into customer behavior and helps businesses prioritize critical variables in decision-making.

The proposed system enables organizations to perform data-driven marketing, personalized product recommendations, customer segmentation, and inventory planning. By comparing the performance and interpretability of both models, the project demonstrates the effectiveness of ensemble learning techniques in solving real-world classification problems. The results indicate that feature importance analysis not only improves model transparency but also assists businesses in understanding customer preferences, leading to better strategic planning and improved business outcomes.

2. INTRODUCTION:

The rapid expansion of e-commerce has transformed the way customers purchase products and services. Every interaction made by users, such as browsing products, adding items to the cart, viewing product details, and making purchases, generates valuable data that can be analyzed to understand customer behavior. With the increasing volume of such data, businesses require intelligent systems capable of predicting customer purchase intentions accurately. Machine Learning provides advanced techniques to analyze historical customer data, identify hidden patterns, and generate reliable predictions that support business decision-making.

Purchase prediction is a supervised Machine Learning classification problem where the objective is to determine whether a customer is likely to purchase a product based on various factors such as browsing history, customer demographics, product category, discounts, previous purchases, session duration, and product ratings. Accurate prediction enables organizations to improve customer experience through personalized recommendations, targeted

advertising, optimized inventory management, and effective marketing campaigns.

This project utilizes two state-of-the-art gradient boosting algorithms, XGBoost (Extreme Gradient Boosting) and LightGBM (Light Gradient Boosting Machine), to build an efficient purchase prediction model. These algorithms are widely recognized for their high prediction accuracy, ability to handle large datasets, and fast training performance. After preprocessing the dataset by handling missing values, encoding categorical features, and selecting relevant attributes, both models are trained and evaluated using standard performance metrics including Accuracy, Precision, Recall, F1-Score, ROC-AUC Score, and Confusion Matrix.

A major focus of this project is Feature Importance Analysis, which identifies the most influential factors affecting customer purchase decisions. Understanding feature importance helps businesses interpret model predictions, improve transparency, and make informed strategic decisions. By comparing the feature importance and predictive performance of XGBoost and LightGBM, this project demonstrates the effectiveness of ensemble Machine Learning techniques in solving real-world e-commerce purchase prediction problems while supporting data-driven business growth.

3. SOFTWARE REQUIREMENTS:

Operating System

Windows 10/11, Linux, or macOS

Programming Language

Python

Development Environment

Google Colab / Visual Studio Code

Libraries and Frameworks

- NumPy
- Pandas
- Scikit-learn
- XGBoost
- LightGBM
- Matplotlib
- Seaborn
- SHAP (for feature importance visualization)

Dataset

- E-Commerce Customer Purchase Dataset (CSV format)

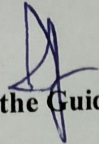
Machine Learning Algorithms Used

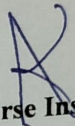
- XGBoost (Extreme Gradient Boosting)
- LightGBM (Light Gradient Boosting Machine)

Evaluation Metrics

- Accuracy
- Precision
- Recall

- F1-Score
- ROC-AUC Score
- Confusion Matrix
- Feature Importance Analysis


Signature of the Guide


Course Instructor